INCH-POUND

MIL-M-71213 (AR)
28 March 1995

MILITARY SPECIFICATION

MOUNT, MACHINE GUN: MK93 MOD1 (DUAL MOUNT)

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Dual Mount, MK93 MOD1, that mounts both the M2 Heavy Barrel Machine Gun (HBMG), .50 caliber, and MK19 Grenade Machine Gun (GMG), 40MM, in both vehicular (M1025/M1026 High Mobility Multipurpose Wheeled Vehicle (HMMWV) and ground mounted modes (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-13855 - Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for

MIL-W-63150 - Weapons and Support Material Standard Quality Assurance Provisions for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-2073 - DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

6650717 - MK93 MOD1

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 6650717, all associated drawings, and with all requirements specified in applicable specifications and drawings, and with requirements specified on Quality Assurance Provisions (QAPs) as well.

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3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Performance characteristic.

3.4.1 MK93 MOD1 assembling: Movement of components. The contractor shall assemble Dual Mounts (Dwg. 6650717) with both M2 HBMG and MK19 GMG and full ammunition cans of .50 caliber M2 dummy linked rounds and of 40MM M922 dummy linked rounds, respectively, to simulate actual operations. All movements (rotation and sliding) shall be free of any binding; in particular the MK175 Adapter with guns (M2 HBMG and MK19 GMG) shall be able to rotate 360 degrees in both directions. Operation of the mechanism shall provide for both bold and accurate operator adjustable controlled traverse and elevation over the full range of elevation and depression of +250 mils (+14 degrees) to -250 mils (-14 degrees). Testing shall be as specified in paragraph 4.8.1.

3.4.2 Functioning. This test shall be performed by the Government at a government test site. With a M2, .50 caliber machine gun and full ammunition box of caliber .50 linked, 4 ball, 1 tracer, ammunition installed and a MK19, 40MM grenade launcher and full ammunition box of 40MM linked M385 TP rounds, the mounts shall be capable of the full range of movements (i.e., 360 degree pintle azimuth traverse each direction, and full gun elevation and depression) without binding or damage to the gun, mount or support. The mounts shall be capable of the full range of movements without binding, and shall not be damaged after being subjected to the firing of not less than 100 rounds each of government standard .50 caliber, linked, 4 ball, 1 tracer ammunition and 40MM M385 TP rounds, respectively. Testing shall be as specified in paragraph 4.8.2.

3.4.3 Interchangeability. Unless otherwise specified on the applicable drawings or in 6.2, all parts shall be interchangeable. Testing shall be as specified in paragraph 4.8.3.

3.5 Workmanship. Workmanship shall be in accordance with MIL-W-63150.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-W-63150 and MIL-W-13855 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
MK93 MOD1	6650717	5

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4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see TABLE I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-M-71213 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
	Mount, Machine Gun: MK93 MOD1 (Dual Mount)			6650717
				NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	MK93 MOD1 Assembly: Movement of Components	ACC 0 REJ 1	3.4.1	4.8.1
	Function Test	ACC 0 REJ 1	3.4.2	4.8.2
	Poor Workmanship	ACC 0 REJ 1	3.5	Visual

NOTES:

AMSMC Form 1570a-E, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sample inspection.

Lot Size	Inspection Level
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	13
91 to 150	13
151 to 280	20

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-M-71213 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Mount, Machine Gun: MK93 MOD1 (Dual Mount)	1 OF 1	6650717
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	MK93 MOD1 Assembling: Movements of Components - MK175 Adapter rotation in hard stand - Rotation of Mount, MG MK93 MOD0 in MK175 MOD0 Adapter - Mechanism Assy, Elevating proper operation when mounted	100%	3.4.1	4.8.1
102	Interchangeability	ACC 0 1/ REJ 1	3.4.3	4.8.3
Minor				
201	Workmanship	ACC 0 REJ 1	3.5	Visual

NOTES:

1/ The three (3) sample units shall be selected for this test at intervals of each 100 mount assemblies produced or every 60 calendar days, whichever comes first.

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4.5 Testing.

4.5.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete mount. If test requirements cited herein are not met, acceptance of the mount shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical, and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c. Examine mounts, partially assembled mounts, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned mount for retest.

4.5.2 Interchangeability test of repair parts. Concurrent repair parts shall be selected and tested for interchangeability (see 3.4.3) as specified in the contract (see 6.2). Concurrent repair parts selected for interchangeability testing shall have met all other examination and test requirements specified herein. When repair parts are tested for interchangeability, the sample parts or assemblies shall be assembled into two mounts which have been randomly selected from an accepted production lot. Complete mounts and mounts used for interchange of concurrent repair parts shall also meet the requirements after interchange of parts, using the test methods specified in 4.8.1 and 4.8.3.

4.6 Inspection for acceptance. The contractor shall perform inspection for acceptance upon completion of all manufacturing and in-process inspection operations, prior to preservation and packing. Inspection for acceptance shall include, but is not limited to, inspection for workmanship, protective finish, missing features (missing operations), characteristics not previously inspected, and characteristics that may have been altered or changed as a result of subsequent operations after prior inspections were completed.

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4.7 Inspection of packaging. Unless otherwise specified (see 6.2), inspection to determine compliance with preservation, packing, and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-STD-2073.

4.8 Methods of inspection. Unless otherwise specified, the contractor's inspection procedures for the assembling test shall be submitted to the contracting officer for review and approval prior to performing first article inspection and the quality conformance inspection.

4.8.1 Mount, Machine Gun: MK93 MOD1 (Dual Mount). The complete mount shall be tested while installed on a suitable test stand (see Figure 1) with both an M2, .50 caliber machine gun and a MK19, 40MM GMG and their full ammunition boxes, respectively. The following test conditions shall apply:

a. The MK175 Adapter shall be able to rotate 360 degrees in both directions without any binding when installed on a suitable test stand.

b. The Mount, MG MK93 MOD0 shall be able to rotate 360 degrees in both directions without any binding when installed with the MK175 Adapter and Mechanism Assembly, Elevating and shall be able to go through a full range of adjustments (i.e., elevation (+14 degrees)/depression (-14 degrees)).

4.8.2 Function test. The test shall be performed at a government site and as indicated on contract. The complete mount shall be tested under the following conditions:

a. Prior to performance of this test, the mount shall have satisfactorily passed the test specified in 4.8.1.

b. The complete mount shall be installed on a suitable test stand (see Figure 1).

c. The mount shall be tested by firing 100 rounds through each gun at nominal (0°) elevation and depression at a single quadrant in azimuth.

d. After function firing, the mount shall be tested for free movement in all directions. It shall also be inspected for any damaged components or welds.

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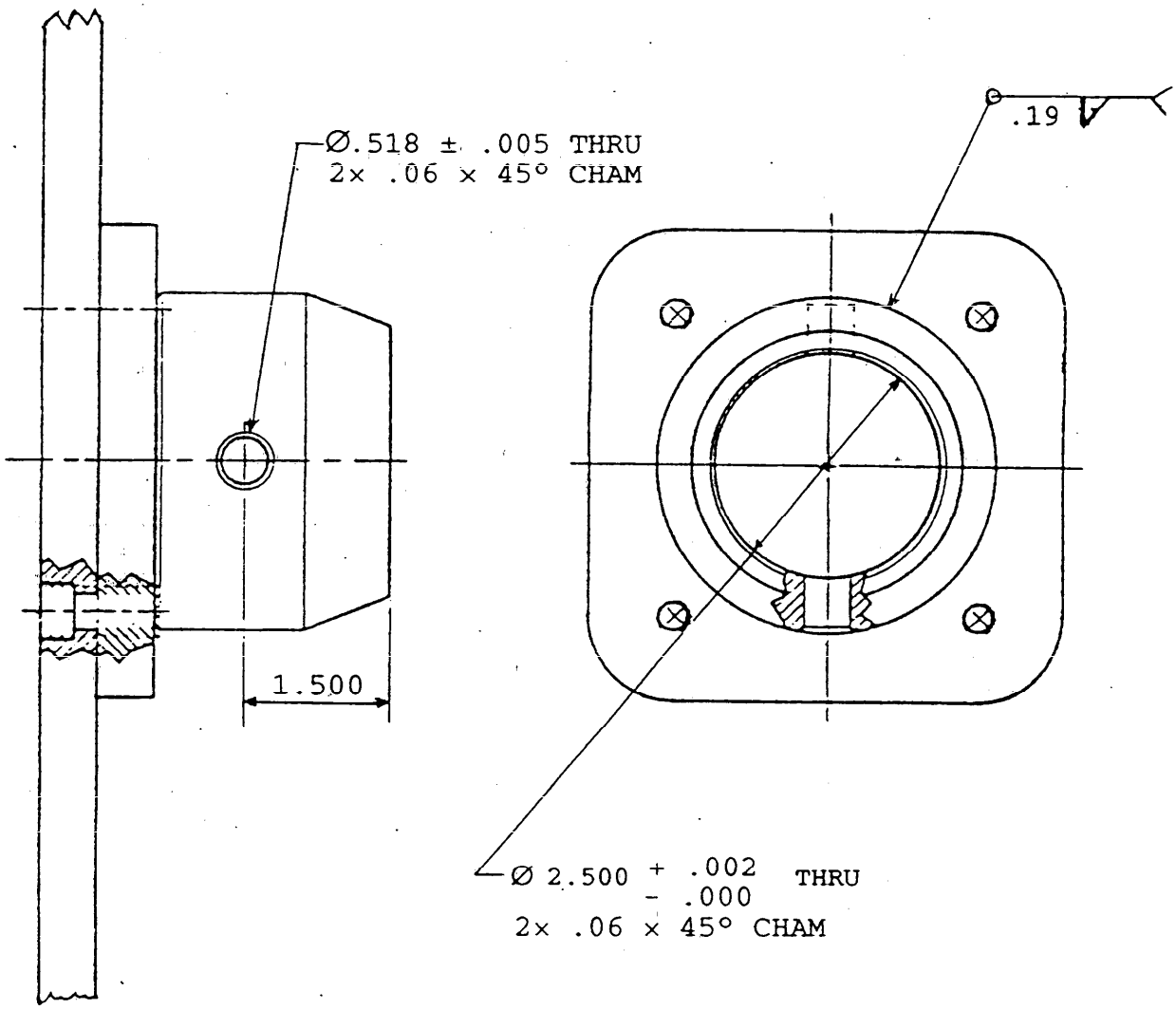


FIGURE 1. Hard stand.

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4.8.3 Interchangeability. Mounts shall be tested for interchangeability by disassembling and then reassembling parts using a prearranged system submitted for review and approval. Interchange of parts shall be accomplished by any convenient method which will insure that no mating parts in reassembled mounts are from the same mount. This shall be accomplished by dividing the parts of each mount assembly in the prearranged system into three groups and distributing the groups into three different trays until each tray contains the parts and assemblies for a complete mount. Observation shall be made for compliance with the applicable requirement. Submission of test plan shall be as indicated on contract.

5. PACKAGING

5.1 Packing. Packing shall be in accordance with SPI-6650717. Level of preservation and level of packing shall be as specified in the acquisition document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mount covered by this specification is intended for use on combat vehicles and ground mounted modes designated on the applicable drawings to properly support and provide for the field use of a Browning machine gun, .50 caliber, M2 (Heavy Barrel-Flexible), and Grenade Machine Gun, 40MM, MK19.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

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h. Criteria for inspection of parts and assemblies (see MIL-W-63150).

i. Interchangeability test instructions (see 3.4.4).

6.3 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

MK19 MODIII Grenade Machine Gun, 40MM
M2 Heavy Barrel Machine Gun, .50 Caliber
M922 Dummy Round, 40MM
M2 Dummy Round, .50 Caliber
M385 Target Practice Round, 40MM

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1095-0280)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-71213	95 MARCH 1995

3. DOCUMENT TITLE: MOUNT, MACHINE GUN: MK93 MOD1 (DUAL MOUNT)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (*Include Zip Code*) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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